

Adversarial client

objective: make the model
misclassify the image

*same optimizer, start image,
perturbation ball, query budget*

Benign client

objective: e.g. restore the
image and keep its label

queries

Classifier

model f behind
a prediction API

queries

outputs

Stateful detector

history of queries;
raises alarms

query-only detectors use the queries;
output-aware detectors use both

```
graph LR; A[Adversarial client] -- queries --> C[Classifier]; B[Benign client] -- queries --> C; C -- queries --> D[Stateful detector]; C -. outputs .-> D;
```